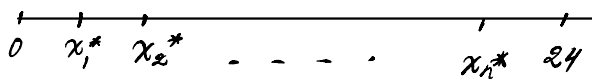
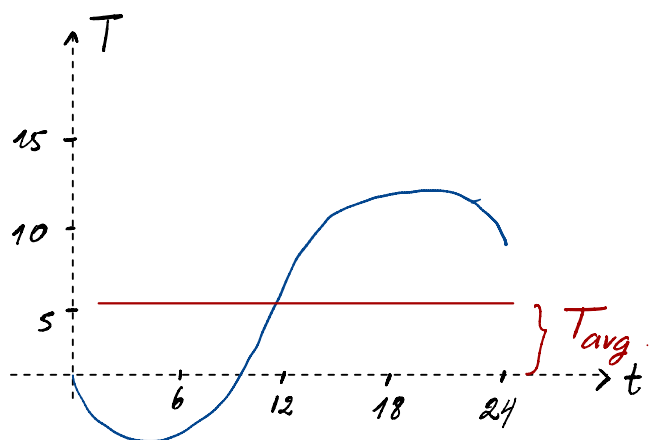


5.5 Average Value of a function.

It is easy to compute the average of finitely many numbers $y_1, y_2, \dots, y_n$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

But, how do we compute the average temperature during a day if infinitely many temperature readings are possible?



then for these numbers

$$\frac{f(x_1^*) + \dots + f(x_n^*)}{n} \quad (†)$$

Notice that if f is defined on $[a, b]$ the number n is given by.

$$n = \frac{b-a}{\Delta x} \quad \left\{ \Delta x = \frac{b-a}{n} \Leftrightarrow n = \frac{b-a}{\Delta x} \right\} \quad \text{So, from } (†).$$

$$\begin{aligned} \frac{f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)}{\frac{b-a}{\Delta x}} &= \frac{1}{b-a} \cdot \left\{ f(x_1^*) + f(x_2^*) + \dots + f(x_n^*) \right\} \Delta x \\ &= \frac{1}{b-a} \cdot \sum_{i=1}^n f(x_i^*) \Delta x. \end{aligned}$$

by taking the limit $n \rightarrow \infty$, thus

$$\bar{f} = \lim_{n \rightarrow \infty} \frac{1}{b-a} \sum_{i=1}^n f(x_i^*) \Delta x = \frac{1}{b-a} \lim_{n \rightarrow \infty} \underbrace{\sum_{i=1}^n f(x_i^*) \Delta x}_{\int_a^b f(x) dx}.$$

Def. The average of f on the interval $[a, b]$ is

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx.$$

Example. Find the average value of the function $f(x) = 1+x^2$ on $[-1, 2]$.

Sln. $a = -1$, $b = 2$.

$$f_{\text{avg}} \stackrel{\text{def}}{=} \frac{1}{b-a} \int_a^b f(x) dx = \frac{1}{2 - (-1)} \int_{-1}^2 (1+x^2) dx = \frac{1}{3} \left(x + \frac{x^3}{3} \right) \Big|_{-1}^2 = 2.$$

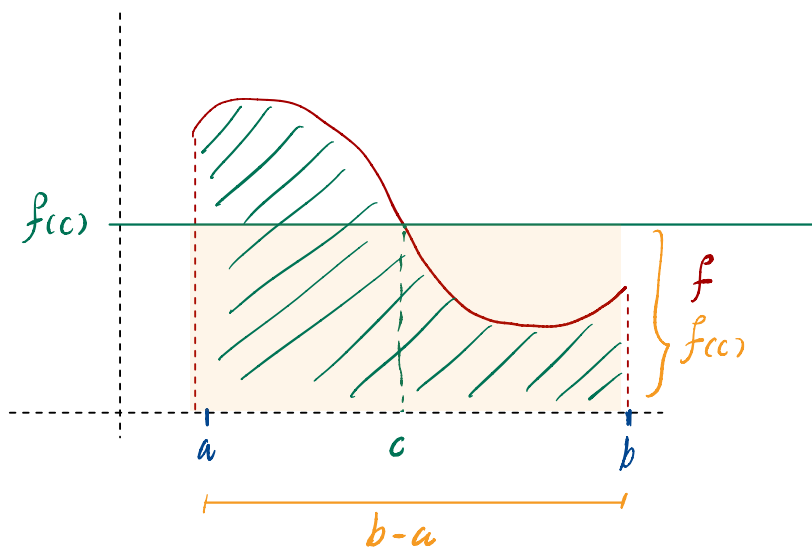
The mean Value thm for integrals. If f is a continuous function on $[a, b]$

then there exists a number $c \in [a, b]$ s.t

$$f(c) = f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx.$$

that is,
$$f(c)(b-a) = \int_a^b f(x) dx.$$

$f(c)(b-a)$ is the area of the rectangle.



Example. Since $f(x) = 1+x^2$ is continuous on $[-1, 2]$ the MVT for integrals says that there is a number $c \in [-1, 2]$ s.t

$$\int_{-1}^2 (1+x^2) dx = f(c)(2 - (-1)).$$

what is c ? We know that $f_{\text{avg}} = 2$. then $f(c) = f_{\text{avg}} = 2$.

So,

$$1+c^2 = 2 \quad \text{so,} \quad c^2 = 1 \quad \text{so,} \quad c = \pm 1. \quad \text{then since}$$

$\pm 1 \in [-1, 2]$ that work in MVT for integrals.

Example. Show that the average velocity of a car over a time interval $[t_1, t_2]$ is the same as the average of its velocities during the trip.

Sln. If $s(t)$ is the displacement of the car at the time t , then by definition of average velocity.

$$\frac{\Delta s}{\Delta t} = \frac{s(t_2) - s(t_1)}{t_2 - t_1}.$$

On the other hand, the average of the velocity function on $[t_1, t_2]$

$$\begin{aligned} \text{is } v_{\text{avg}} &= \frac{1}{b-a} \int_a^b v(t) dt. \quad b=t_2, \quad a=t_1 \\ &= \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} v(t) dt. \\ &= \frac{1}{t_2 - t_1} s(t) \Big|_{t_1}^{t_2} \quad \text{by FTC Part 2.} \\ &= \frac{s(t_2) - s(t_1)}{t_2 - t_1} = \frac{\Delta s}{\Delta t}. \end{aligned}$$